Professional Online Academy Vu Topper RM

MTH-601 Operations Research
Update MCQ'S Final Term
By Vu Topper RM

Follow Us on YouTube:
@vutopperrm

For More Help Contact What's app Me!!!
Number's = # 0322-4021365, 0316-4980094

Rizwan Manzoor
Vu Topper RM



بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:1 **(Marks:1)** **Vu-Topper RM**

How is Vogel's Approximation Method (VAM) different from the least cost method (LCM)?

A. VAM does not consider penalties
B. VAM considers penalties for wrong allocations
C. VAM does not consider unit costs
D. VAM starts from the bottom right corner

Question No:2 **(Marks:1)** **Vu-Topper RM**

The Transportation Problem is one of the sub-class of ----------.

A. Inventory Control Problem
B. Stochastic Problem
C. Network Flow Problem
D. Linear Programming Problem

Question No:3 **(Marks:1)** **Vu-Topper RM**

If the cost matrix in an Assignment problem is not square then which the following modification will be made to balance the given problem?

A. Add a dummy row(column) with arbitrary(including negative or positive) cost elements
B. Add a dummy row(column) with positive cost elements
C. Add a dummy row(column) with zero cost elements
D. Add a dummy row(column) with negative cost elements

Question No:4 **(Marks:1)** **Vu-Topper RM**

Suppose that a taxi firm has four taxis (the agents) available, and four customers (the tasks) wishing to be picked up as soon as possible. The firm prides itself on speedy pickups, so for each taxi the cost of picking up a particular customer will depend on the time taken for the taxi to reach the pickup point. This is _____ problem.

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

A. transportation
B. unbalanced assignment
C. balanced assignment
D. linear programming

Question No:5 **(Marks:1)** **Vu-Topper RM**

_____ problems are _____ problems

A. Assignment, combinatorial
B. transportation, unbalanced
C. none of these
D. linear programming, simplex

Question No:6 **(Marks:1)** **Vu-Topper RM**

For calculating the penalty cost of a row in Vogel's approximation method, if the smallest element is '3' and other elements are 3, 4, 5 then which of the following will be taken as next highest?

A. 3
B. 5
C. 7
D. 4

Question No:7 **(Marks:1)** **Vu-Topper RM**

To balance Assignment matrix we have to----------.

A. open a Dummy column
B. you cannot balance the assignment matrix
C. open a Dummy row
D. open either a dummy row or column depending on the situation

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:8 **(Marks:1)** **Vu-Topper RM**

Under which of the following condition in the North West Corner Method, after first allocation in cell (1,1), next allocation along the diagonal in cell (2,2) can be made?

A. If in the first row and first column, each resource and sink contain equal units

B. No such condition exist

C. If Total Supply is not equal to Total Demand

D. Total Supply = Total Demand

Question No:9 **(Marks:1)** **Vu-Topper RM**

In North West Corner Rule, we start initially from which of the following cell in the standard transportation table?

A. Upper left

B. Upper right

C. Upper middle

D. From any arbitrary cell in first row

Question No:10 **(Marks:1)** **Vu-Topper RM**

In transportation models, what does the decision variable xij typically represent?

A. The total production capacity at factory i.

B. The total demand at warehouse j.

C. The quantity of goods transported from factory i to warehouse j.

D. The cost of transporting one unit from factory i to warehouse j.

Question No:11 **(Marks:1)** **Vu-Topper RM**

If and are row and column numbers respectively, then the implied cost is given by: Ui Vj Cij

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

A. Ui– Vj
B. Ui/Vj
C. Ui + Vj
D. Ui × Vj

Question No:12 **(Marks:1)** **Vu-Topper RM**

For North West Corner method, in the first row and first column, each resource and sink contain '5' units; then after allocating the appropriate amount 'x11' in the cell (1,1), in which of the following cell the next allocation will be zero?

A. Neither in (1,2) nor in (2,1) but in along the diagonal in (2,2)
B. In cell (2,1) only
C. Arbitrarily chosen either in (1,2) or in (2,1)
D. In cell (1,2) only

Question No:13 **(Marks:1)** **Vu-Topper RM**

Which of the following reason to prefer Vogel's approximation method over other two methods?

A. Vogel's method gives the shortest transportation rout
B. Vogel's method allocate minimum amount to the variable with least unit cost
C. Vogel's method gives optimal time plan of transportation
D. Vogel's method gives the feasible solution near to the optimal

Question No:14 **(Marks:1)** **Vu-Topper RM**

The amounts shipped from a dummy source represent shortages at the receiving destinations

True
False

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

Question No:15 **(Marks:1)** **Vu-Topper RM**

In the Least Cost Method, for any cell both demand and supply are satisfied then which of the following will be crossed out?

A. Only column
B. Either of row or column
C. Only row
D. Both row and column

Question No:16 **(Marks:1)** **Vu-Topper RM**

The similarity between Assignment Problem and Transportation problem is----------.

A. both are square matrices
B. both are rectangular matrices
C. both have objective function and non negativity constraints
D. both can be solved by graphical method

Question No:17 **(Marks:1)** **Vu-Topper RM**

While solving an Assignment problem by Hungarian's method, the horizontal and vertical lines drawn to cover all zeros of modified cost matrix must be ---------.

A. must be equal to m.n (where m and n are number of rows and columns)
B. equal to number of rows or columns
C. equal to each other
D. m + n(m and n are number of rows and columns)

Question No:18 **(Marks:1)** **Vu-Topper RM**

If a balanced Transportation problem with 'm' sources and 'n' sinks then degeneracy will arise only if there are less than ---------- independent allocations in the solution.

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

A. m–n–4
B. m–n–2
C. m–n–3
D. m–n–1

Question No:19 (Marks:1) Vu-Topper RM

In an Assignment problem, while applying Hungarian's method, if in the final modified matrix any row or column does not have single zero, then which of the following is true?

A. Unique assignment will be made
B. None of the above
C. No assignment can be made
D. Multiple assignments can be made

Question No:20 (Marks:1) Vu-Topper RM

What is done with the row/column after an allocation is made in the LCM?

A. Eliminate the row/column
B. Adjust the demand at the destination
C. Increase the cost of transportation
D. Retain the row/column for future allocations

Question No:21 (Marks:1) Vu-Topper RM

If the total cost of the assignment for all the tasks is equal to the sum of the costs for each agent, then the problem is called _____.

A. optimization
B. simplex
C. linear assignment
D. unbalanced assignment

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

Question No:22 **(Marks:1)** **Vu-Topper RM**

In the context of transportation problems, what does it mean for a solution to be non-degenerate?

A. It has no allocations at all.
B. It has the minimum possible number of allocations.
C. It has the maximum possible number of allocations.
D. It has exactly (m + n - 1) allocations in independent positions.

Question No:23 **(Marks:1)** **Vu-Topper RM**

For an unbalanced Transportation problem, if the total demand is MORE than total supply then which of the following is true in order to balance the problem?

A. One constraint will have evacuate
B. One constraint will have to add
C. A dummy source would have to include with supply equal to shortage
D. A dummy sink would have to include with demand equal to the surplus

Question No:24 **(Marks:1)** **Vu-Topper RM**

There always exists a non-degenerate Feasible solution of a Balanced Transportation Problem.

True
False

Question No:25 **(Marks:1)** **Vu-Topper RM**

If a balanced Transportation problem with '7' sources and '6'sinks then its non-degenerate basic feasible solution will have---------- independent allocations.

A. 11

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

B. 12
C. 13
D. 10

Question No:26 **(Marks:1)** **Vu-Topper RM**

Despite of minimizing the cost of Transportation, if we want to maximize the profit then which of the following modification would have to do?

A. Subtract '1' from all the cost coefficients in Transportation matrix
B. Divide all the cost coefficients in Transportation matrix by minimum cost coefficient
C. Add '1' from all the cost coefficients in Transportation matrix
D. Multiply '–1' to all the cost coefficients in Transportation matrix

Question No:27 **(Marks:1)** **Vu-Topper RM**

If any or all of the artificial variables do not leave the basis in the final solution, then this indicates that the problem _____ .

A. non of these
B. does not have a solution
C. may have a solution
D. must have a solution

Question No:28 **(Marks:1)** **Vu-Topper RM**

Once an LP problem is converted to a table, then calculation steps would be same for _____ method(s).

A. Big M
B. all of above
C. simplex
D. Two phase

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

VuTopperRM

Question No:29 **(Marks:1)** **Vu-Topper RM**

In two phase method, for the phase-I, if the objective function in terms of artificial variables is not minimized then the given problem has---------.

A. Degeneracy
B. Feasible solution
C. Optimal solution
D. Infeasible solution

Question No:30 **(Marks:1)** **Vu-Topper RM**

The role of artificial variables in the 'Big M' technique is to _____.

A. maximize the objective function
B. minimize the objective function
C. ensure feasibility of the initial basic solution
D. optimize resource allocation

Question No:31 **(Marks:1)** **Vu-Topper RM**

By Simplex method, to maximize 'Z = 2x–9y'of an LP problem, if 'z=0' for the initial iteration then for its next improved solution, which of the following would
be the next entering variable?

A. y<0
B. x>0
C. x<0
D. y>0

Question No:32 **(Marks:1)** **Vu-Topper RM**

If a variable in the Primal is unrestricted in sign, then the corresponding constraint in the dual will be of -------- type and vice versa.

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

A. =
B. none of these
C. <=
D. >=

Question No:33 **(Marks:1)** **Vu-Topper RM**
While solving an LP problem by Simplex method, the inclusion of slacks in the constraints' inequalities helps in finding ---------- variables.

A. basic
B. non-basic
C. artificial
D. decision

Question No:34 **(Marks:1)** **Vu-Topper RM**
In two phase method, for the phase-I, the problem has infeasible solution if the minimum value of objective function ----------.

A. is imaginary
B. greater than zero
C. less than zero
D. equal to zero

Question No:35 **(Marks:1)** **Vu-Topper RM**
In the Two Phase Method the objective of Phase 2 is to _____.

A. maximize the artificial variables
B. find the optimum basic solution for the original problem
C. minimize the original objective function
D. maximize the original objective function

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:36 **(Marks:1)** **Vu-Topper RM**

Dual of a Dual of Dual is------.

A. Single dual
B. Dual
C. Double dual
D. Primal

Question No:37 **(Marks:1)** **Vu-Topper RM**

In Big M-method, if the objective function is expressed in terms of non-basic variable then the coefficient of artificial variable in this iteration must be ----
---.

A. one
B. M
C. –M
D. zero

Question No:38 **(Marks:1)** **Vu-Topper RM**

If in a LP problem, the objective is to maximize: z = 2x+3y, with all associated constraints of '=<' type, then in the 1st iteration, z = ---------.

A. 0
B. 5
C. 6
D. 1

Question No:39 **(Marks:1)** **Vu-Topper RM**

The insensitivity of the solution relative to the original decision variables in an LP problem which is solved by M-method is overcome by ----------.

A. Graphical method

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

B. Two phase method
C. Duality principle
D. Simplex method

Question No:40 **(Marks:1)** **Vu-Topper RM**
Simplex method can only be applied to problems with two decision variables.
True
False

Question No:41 **(Marks:1)** **Vu-Topper RM**
If the optimal table contains more zero coefficients in the objective row than the number of variables in the basis in simplex procedure then the solution
will _____.
A. not exist
B. indicate an unbounded solution
C. suggest another optimal basic feasible solution
D. become infeasible

Question No:42 **(Marks:1)** **Vu-Topper RM**
Which of the following is true about an LP problem which is solved by M-method and having infeasible solution?
A. Final iteration table include one of the artificial variables.
B. All choices are equivalent.
C. Penalty 'M' may not able to force artificial variable to zero in final iteration.
D. Associated constraints are not consistent

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:43 **(Marks:1)** **Vu-Topper RM**

If the primal has an unbounded solution, then the dual has------

A. infeasible solution
B. optimal solution
C. bounded solution
D. no solution

Question No:44 **(Marks:1)** **Vu-Topper RM**

Dual of a Dual is ------

A. (Dual)^2
B. Dual
C. Primal
D. Primal-du

Question No:45 **(Marks:1)** **Vu-Topper RM**

In an LP problem, to evaluate the basic variables in Simplex method, which of the following method is used?

A. Gauss Elimination method
B. Gauss-Jordan method
C. Gauss-Seidel method
D. Gauss Least Square method

Question No:46 **(Marks:1)** **Vu-Topper RM**

In two phase method process, first phase _____ the sum of artificial variables.

A. maximize
B. non of these
C. minimize
D. maximize or minimize depending on the situation

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

Question No:47 **(Marks:1)** **Vu-Topper RM**

If in the primal problem , there are 2 variables and 3 constraints and the elements of the right hand side of the constraints are 200,567,987 respectively, then its dual objective function is --------.

A. Z = 200y1 + 567y2

B. Z = 200y1 + 567y2 + 987y3

C. Z = y1 + y2 + y3

D. Z = y1 + 3y3

Question No:48 **(Marks:1)** **Vu-Topper RM**

In the initial iteration of Big M-method, the artificial variables appear in ---------.

Non-basic variables' set

Bas

Question No:49 **(Marks:1)** **Vu-Topper RM**

In the final iteration of M-method to obtain the optimal solution, the artificial variable must------- the basis.

A. include in

B. leave

C. add in each variable of

D. subtract from each variable o

Question No:50 **(Marks:1)** **Vu-Topper RM**

A Linear Programming problem may not have any feasible solution.

False

True

Vu Topper RM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Question No:51 **(Marks:1)** **Vu-Topper RM**

Minimizing Z = 4x1 + 2x2 + x3 is equivalent to max imizing Z = _____.

A. −4x1 − 2x2 − x3

B. −4x1 + 2x2 − x3

C. −4x1 + 2x2 + x3

D. −4x1 − 2x2 + x3

Question No:52 **(Marks:1)** **Vu-Topper RM**

In two phase method, for the phase-I, if the given problem has feasible solution then----------.

A. both objective function and artificial are zero

B. objective function is arbitrary but artificial is zero

C. objective function is zero but artificial may arbitrary

D. both objective and artificial can have arbitrary values

Question No:53 **(Marks:1)** **Vu-Topper RM**

When _____ fail to provide an initial basic feasible solution in the simplex method, we use the Big M Method.

A. slack variables

B. pivot variable

C. artificial variables

D. objective function

Question No:54 **(Marks:1)** **Vu-Topper RM**

When constraints are of greater than or equal to type, then we _____.

A. non of these

B. both of these

C. add slack variable and subtract artificial varialbe

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

D. subtract slack variable and add artificial variable

Question No:55 **(Marks:1)** **Vu-Topper RM**

A linear programming problem _____ optimal solutions.

A. must have the multiple
B. may not have the multiple
C. must not have the multiple
D. may have the multiple

Question No:56 **(Marks:1)** **Vu-Topper RM**

If an artificial variable appears in the basis of optimal table in linear programming, what does this indicate about the feasibility of the problem?

A. Feasible optimal solution is bounded
B. Feasible solution is degenerate
C. No feasible optimal solution exists
D. Feasible solution is unbounded

Question No:57 **(Marks:1)** **Vu-Topper RM**

While solving an LP problem, if the candidate for a leaving variable becomes zero, then solution is said to be --------.

A. degenerate solution
B. non-degenerate solution
C. infeasible solution
D. optimal solution

Question No:58 **(Marks:1)** **Vu-Topper RM**

For finding the maximum profit in an enterprise of selling two products such that 'freezing' the sale of one product and keep selling the other. This scenario is studied under ----------.

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

Vu Topper RM

A. Duality
B. Artificial variable technique
C. Degeneracy
D. Un-boundedness

Question No:59 **(Marks:1)** **Vu-Topper RM**
In two phase method, for the phase-I, a new objective function is expressed as -----.
A. difference of slack variables
B. sum of artificial variables
C. sum of slack variables
D. difference of artificial variables

Question No:60 **(Marks:1)** **Vu-Topper RM**
Primal of a dual is-----.
A. Prime primal
B. Primal
C. Prime dual
D. Dual

Question No:61 **(Marks:1)** **Vu-Topper RM**
In Simplex method, which of the following is the standard equality corresponding to the constraint"3x+5y>=2"?
A. 3x+2y–S+A=2
B. 3x+2y–S–A=2
C. 3x+2y+S+A=2
D. 3x+2y+S–A=2

Question No:62 **(Marks:1)** **Vu-Topper RM**
When we have type constraints, then we convert it to equality constraints by introducing _____ variable(s) and solve it by _____ method.
A. artificial, Two phase

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365

VuTopperRM

B. artificial, Big M
C. slack, Big M
D. artificial and slack, Big M or Two phase

Question No:63 (Marks:1) Vu-Topper RM
In the Simplex method to solve an LP problem, Gauss Jordan Elimination method demands that all the key column's entries should be --------- except key row(pivot) entry.

A. zero
B. maximum or minimum
C. strictly negative
D. strictly positive

Question No:64 (Marks:1) Vu-Topper RM
If Primal Problem is a maximization problem, then the dual will be ------

A. mixed Problem
B. maximization Problem
C. none of the above
D. minimization Problem

Topper RM

Visit My YouTube Channel
For Subjective and More Important Files
Channel Name = #VuTopperRM

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

For More Help Contact What's app 03224021365